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Group 13

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GitHub Link: `https://github.com/esthxrchxng/CDSProj.git`

Contents

1 Project Aim **4**
Problem Statement . . . 4
Objective . . . 4

2 Dataset and Collection: **4**
2.1 Bloomberg Datasets . . . 4
2.2 Supplementary Datasets . . . 5
2.2.1 Reddit . . . 5

3 Data Pre-processing: **6**
3.1 Bloomberg Terminal . . . 6
3.1.1 Bloomberg File Parsing and Column Pruning . . . 6
3.1.2 Internal File Merges . . . 6
3.1.3 Master Calendar Alignment . . . 6
3.1.4 Missing Values Treatment . . . 7
3.1.5 Handling the WTI Crude Oil Negative Price Event . . . 7
3.1.6 Feature Engineering . . . 7
3.1.7 Redundant Feature Removal . . . 8
3.1.8 Rolling Window Warm Up and Final Row Removal . . . 8
3.1.9 Train-Test Split and Feature Scaling . . . 8
3.2 Reddit . . . 9
3.2.1 Enhanced Data Preprocessing and Ticker Extraction . . . 9
3.2.2 Semantic Feature Engineering and Weighting Strategy . . . 9
3.2.3 Model Architecture and Hyperparameter Optimization . . . 9
3.2.4 Stratified Data Partitioning and Train-Test Set Split . . . 9
3.2.5 Probabilistic Inference (Softmax Logic) and Metadata Re-linking . 10
3.2.6 Multi-Level Aggregation and Ticker-Specific Grouping . . . 10
3.2.7 Time-Series Smoothing and Momentum Signal Generation . . . 10
3.2.8 Sentiment Evaluation and Discussion . . . 11
3.2.9 Real-Time Data Retrieval and Temporal Extension . . . 11

4 Problem and Algorithm/Model: **11**
4.1 Problem Definition . . . 11
4.2 Algorithm/Model Initialization . . . 12
4.3 Modelling Architecture . . . 13

5 Evaluation Methodology: **13**
5.1 Data Usage & Chronological Splitting Design . . . 13
5.2 TimeSeriesSplit & OOF bridge generation . . . 14
5.3 Sentiment Alignment, Aggregation and Missing-value Handling . . . 14
5.4 Evaluation Metrics and Thresholding Strategy . . . 15

6 Results and Discussion: **15**
6.1 Results . . . 15
6.1.1 Financial-only Results & Precision Ceiling . . . 15
6.1.2 Final Stacked-Model Performance on the Blind 2022-2025 Test . . . 17
6.1.3 Interpretation of the Meta Learner . . . 17

6.1.4 Predictions for Jan - Apr 2026 18
6.2 GUI 18
6.2.1 Public Dashboard: S&P 500 Downturn Visualisation 18
6.2.2 Admin Dashboard: Model Performance and Social Analytics 22

7 Appendix **24**
7.1 Datasets 24
7.2 Models/Algorithms 36

8 References **40**

9 Sources **41**

1. Project Aim

Problem Statement

How might we predict potential downturns in the S&P 500 and provide early alerts to retail investors so they can reduce potential losses?

Objective

To develop a predictive system that leverages social media sentiment data and market financial data to identify a potential downturn in the S&P 500.

The goal is to provide individual retail investors with timely alerts indicating increased probability of a bear market trend, enabling them to make informed decisions and mitigate potential losses.

2. Dataset and Collection:

This project utilizes a combination of financial market datasets obtained from Bloomberg Terminal and Reddit. The datasets consist of historical time-series data related to equity market performance, market volatility, interest rates, and credit market risk. These datasets were selected because they represent key financial indicators that are commonly associated with market regime changes and potential downturns.

2.1 Bloomberg Datasets

All financial data used in this project were sourced from the Bloomberg Terminal. Each dataset was exported directly from Bloomberg as an individual Excel file. Bloomberg's export format embeds several metadata rows above the actual tabular data, so a custom Python parsing function was implemented to automatically detect the true header row (which was identified as the first row whose leftmost cell contains the string "Date") before loading the data. This ensured consistent and reliable ingestion across all 17 raw files without manual editing of any source file.

The 17 raw files collectively span six asset classes: U.S. equity indices, equity index futures, implied volatility, government bond yields, precious metals, and crude oil. Together they cover the period from 2 April 1990 to 2 April 2026, representing approximately 36 years of daily market observations. Table 1 below summarises each instrument, the Bloomberg fields extracted, and its effective date range.

Table 1. Bloomberg Extracted Dataset

Asset Class	**Instrument**	**Bloomberg Fields**	**Date Range**	**Raw Rows**
U.S. Equity Index	S&P 500 — Price (SPX)	Last, Open, High, Low	Apr 1990 – Apr 2026	9,067
U.S. Equity Index	S&P 500 — Volume (SPX)	Volume	Apr 1990 – Apr 2026	9,081
Implied Volatility	CBOE VIX Index	High, Low	Jan 1992 – Apr 2026	8,651
U.S. Treasury Yields	10-Year Yield (USGG10YR)	Last	Apr 1990 – Apr 2026	9,380
U.S. Treasury Yields	2-Year Yield (USGG2YR)	Last	Apr 1990 – Apr 2026	9,380
U.S. Treasury Yields	3-Month Yield (USGG3M)	Last	Apr 1990 – Apr 2026	9,382
Precious Metals	Gold Spot (XAU/USD)	Last	Apr 1990 – Apr 2026	9,338
Peer Equity Index	Dow Jones Industrial Average	Last, Open, High, Low	Apr 1990 – Apr 2026	9,067–9,068
Peer Equity Index	Nasdaq Composite (NDX)	Last, Open, High, Low	Apr 1990 – Apr 2026	9,067
Global Equity Index	MSCI World (MXWO)	Last, Open, High, Low	Apr 1990 – Apr 2026	9,389–9,394
Equity Index Futures	E-mini S&P 500 Front Contract (ES1)	High, Low	Sep 1997 – Apr 2026	7,246
Equity Index Futures	E-mini S&P 500 2nd Contract (ES2)	High, Low	Sep 1997 – Apr 2026	7,195
Crude Oil	WTI Crude Front Contract (CL1)	Last	Apr 1990 – Apr 2026	9,065
Crude Oil	Brent Crude Front Contract (CO1)	Last	Apr 1990 – Apr 2026	9,209

Two instruments have shorter histories than the rest. The CBOE VIX Index with reliable daily data beginning in January 1992, and the E-mini S&P 500 futures contracts (ES1 and ES2), which were listed in September 1997. Features derived from these instruments are therefore unavailable for the earliest portion of the sample, though this does not affect the usable modelling period once the rolling warm-up window has been observed.

2.2 Supplementary Datasets

2.2.1 Reddit

The dataset "Reddit Finance Posts Dataset SP500" sourced from HuggingFace contains 431,923 Reddit Posts from 2008 to July 2025, collected from 20 finance-related subreddits via the Reddit API (refer to Table 3 in appendix). The S&P 500's quantitative financial

analysis is enhanced by this dataset. Other data streams are rapidly influencing market equilibrium in today's financial environment. In order to measure retail investor sentiment, this project incorporates unstructured social conversation from the Reddit financial ecosystem, which includes subreddits like /r/stocks and /r/wallstreetbets.

3. Data Pre-processing:

3.1 Bloomberg Terminal

Raw Bloomberg exports required several structured pre-processing steps before any features could be engineered. These steps are described in the order they were applied in the pipeline: file-level parsing, internal merging, column standardisation, master-calendar alignment, missing value treatment, and finally post-feature-engineering cleaning steps including redundant feature removal and feature scaling.

3.1.1 Bloomberg File Parsing and Column Pruning

As Bloomberg exports embed metadata rows above the data table, each file was loaded in two passes. First without a header to locate the row containing the string "Date", then re-loaded with that row set as the header. Following this, any columns with auto-generated "Unnamed" labels which were artefacts of Bloomberg's multi-column export format were dropped automatically.

Certain Bloomberg fields that were present in the raw exports but not relevant to the modelling task were also removed at this stage. Specifically, the `PX_BID` field appeared in several files (notably the crude oil exports) and was discarded, retaining only `PX_LAST` for price-based instruments. This column pruning step ensured that the merged dataset contained only theoretically motivated variables.

3.1.2 Internal File Merges

Bloomberg's per-series export constraints meant that some instruments were split across multiple files. Before constructing the master dataset, these were recombined using inner joins on the Date column:

- S&P 500: OHLC prices (from `SPX_Last_Open_val.xlsx`) were joined with daily volume (from `SPX_Volume.xlsx`).
- Dow Jones, Nasdaq, and MSCI World: Last/Open price files were joined with their respective High/Low files, producing a full OHLC record for each index.

Following these internal merges, all column names were standardised using asset-specific prefixes (for example, `PX_LAST` in the SPX file became `SPX_PX_LAST`, and `PX_HIGH` in the VIX file became `VIX_PX_HIGH`). This naming convention prevented column conflicts during the subsequent cross-asset merge and made the source of each feature unambiguous throughout the pipeline.

3.1.3 Master Calendar Alignment

All instruments were aligned to the S&P 500 trading calendar using left joins on the Date column, with SPX serving as the anchor. This design choice is consistent with the predic-

tion objective: since the target variable is defined on SPX closing prices, it is appropriate for the observation set to correspond exactly to SPX trading days. After converting all Date columns to pandas datetime objects and sorting each file chronologically, the 12 auxiliary datasets were sequentially left-merged onto the SPX base. The resulting raw merged panel contained 9,067 rows and 30 columns.

3.1.4 Missing Values Treatment

The left-join structure introduced missing values wherever an auxiliary instrument did not have a record on a given SPX trading day, arising from differing holiday calendars, earlier start dates for some instruments, or gaps in Bloomberg's coverage. These were handled using two sequential fill strategies.

Forward-fill (`ffill`) was applied first across all auxiliary columns, carrying the most recent available observation forward to fill gaps caused by non-overlapping trading calendars. This is the standard convention for financial time series, as it reflects the last known market price and does not introduce any look-ahead bias.

Backward-fill (`bfill`) was then applied to resolve any remaining leading NaN values that forward-fill cannot address — specifically, rows at the very beginning of the series where no prior observation exists. This affected primarily the VIX columns, which only begin in January 1992 but needed to be populated for SPX trading days back to April 1990 during the initial warm-up phase.

Following these two steps, the merged dataset contained zero missing values across all 30 columns, confirmed by a post-merge null check.

3.1.5 Handling the WTI Crude Oil Negative Price Event

A known anomaly in the WTI crude oil series is the negative settlement price recorded on 20 April 2020, when front-month futures briefly traded below zero due to storage constraints during the COVID-19 pandemic. Since log-return calculations require strictly positive values, WTI and Brent prices were clipped to a minimum floor of $0.01 per barrel before any logarithmic transformations were applied to the oil series. This avoids undefined values without meaningfully distorting the historical return distribution. This adjustment was applied prior to feature engineering so that all oil-derived return features were computed on the corrected series.

3.1.6 Feature Engineering

Following data cleaning, 111 features were engineered from the cleaned multi-asset panel across six thematic groups:

- SPX price, return and volatility features
- technical indicators
- VIX-derived features
- Treasury yield curve features
- gold and oil features
- cross-index breadth and divergence features.

A composite multi-asset stress score was additionally constructed by summing five binary stress flags. All rolling computations use exponentially spaced windows of 5, 10, 21, 63, 126, and 252 days, corresponding approximately to one week, two weeks, one month, one quarter, six months, and one year respectively. To avoid look-ahead bias, all rolling and lag-based features at row t are computed exclusively from price and market data up to and including day t, while the target variable is defined over the forward window beginning at day $t + 1$.

Table 2 in the appendix provides a consolidated reference of all 111 engineered features, their computation methods, and the rationale for their inclusion.

3.1.7 Redundant Feature Removal

The feature engineering step described in Section 3.1.6 produced 140 columns in total. A pairwise absolute correlation filter was applied to eliminate redundant features. Any feature whose absolute Pearson correlation with any other earlier feature exceeded 0.97 was dropped. This threshold removed 28 features, comprising primarily the raw price-level series for peer indices (such as `DJ_PX_LAST`, `NDX_PX_LAST`, `MXWO_PX_LAST`) and their OHLC components, which were nearly perfectly collinear with the engineered return and ratio features derived from them. Other dropped columns included intermediate calculation columns such as `SPX_MA_50` and `SPX_MA_200`, which were redundant given the distance-from-MA features computed from them, along with `VIX_PX_LOW`, `CO1_PX_LAST`, `US2Y_MA_20`, and the raw ES1 and ES2 price columns.

3.1.8 Rolling Window Warm Up and Final Row Removal

Feature engineering introduced leading NaN values in proportion to each feature's maximum rolling window. The longest window used was 252 trading days (one calendar year), employed in the 252-day rolling maximum drawdown and rolling percentile rank features. After applying `dropna()` to remove all rows containing any NaN, 272 rows were removed from the front of the dataset, shifting the effective start date from April 1990 to 1 April 1991. The final feature-engineered dataset therefore spans April 1991 to April 2026 with 8,795 complete observations and zero missing values.

An additional 30 rows were removed from the end of the dataset, corresponding to observations for which the forward-looking 30-day target variable cannot be computed because the future price window extends beyond the available data. This yielded a final modelling dataset of approximately 8,765 observations.

3.1.9 Train-Test Split and Feature Scaling

The dataset was split chronologically into a training set comprising the earliest 80% of observations and a test set comprising the remaining 20%, producing a training set of 7,012 rows and a test set of 1,753 rows. Random shuffling was deliberately avoided, as doing so would contaminate future information into the training set and produce artificially optimistic performance estimates.

Feature scaling was performed using `StandardScaler`, which standardises each feature to zero mean and unit variance. Critically, the scaler was fitted exclusively on the training set and then applied to transform both the training and test sets. Fitting the scaler on

the combined dataset would constitute look-ahead bias, as it would expose information about the test period's distributional properties during training.

3.2 Reddit

3.2.1 Enhanced Data Preprocessing and Ticker Extraction

To ensure that the project remains relevant to the present S&P 500 constituents, an Automated Ticker Acquisition technique was implemented. To ensure pipeline resilience, the top 70 most traded stocks were hardcoded as a backup. S&P 500 constituent list was dynamically retrieved from reliable sources (GitHub/DataHub). The Heuristic Extraction Logic is the essential component of this stage.

The system recognizes both "cashtags" (like `$NVDA`) and isolated uppercase strings (like `AAPL`) in post titles and content using Regular Expressions (Regex). A Validation Layer compares these matches to the S&P 500 list reducing the high noise-to-signal ratio of social media. This keeps typical linguistic noise like "A," "FOR," and "I" from being mistakenly identified as tickers. Three crucial metrics are produced by this process: a `ticker_count` to gauge the depth of asset conversation within a single thread, a `has_ticker` boolean flag for dataset pruning, and a unique list of tickers each post.

3.2.2 Semantic Feature Engineering and Weighting Strategy

NLP pipeline is required to convert unpredictable, slang-heavy social media talk into a structured financial signal. Semantic feature engineering was used before model ingestion to reduce the high noise-to-signal ratio. Weighted Polarity Formula was used, giving a title:test weight ratio of 60:40, as titles express the strongest sentiment. Semantic feature engineering was carried out by adding context indicators like `[TICKERS: XYZ]` to the text in order to improve the model's comprehension of financial subtlety. Title-Boosting technique is used to increase the title string's "Attention" weight during the Transformer's forward pass by duplicating it within the input sequence. Class Balancing is used to overcome the inherent optimism of retail social media, sampling an equal number of positive and negative messages to prevent the model from developing a "Bullish Bias."

3.2.3 Model Architecture and Hyperparameter Optimization

The analytical engine is built upon DistilBERT, a Transformer model optimized for high-performance inference. DistilBERT was chosen because it operated with a far smaller computing footprint while maintaining 97% of BERT's linguistic nuance.

To address the peculiar "slang" of retail investors, regularization approaches are applied during the Supervised Fine-Tuning phase. To avoid overfitting on social media noise, the Dropout is set to 0.3. Weighted Cross-Entropy Loss function is made as "Bearish" signals are statistically less common in retail datasets, it imposes a bigger mathematical penalty for misclassifying them, which forces the model to become more sensitive to downside risk indicators.

3.2.4 Stratified Data Partitioning and Train-Test Set Split

After the DistilBERT architecture is trained, a strict data partitioning method is put in place to make sure the model's predictive power generalizes to unforeseen market

situations. To prevent the model from developing a "majority-class bias" (also known as optimism bias in finance), the dataset was balanced using an under-sampling and capping technique. Sentiment in unprocessed social media data is often biased toward positive "hype," which may cause a model to only estimate the most common label rather than picking up on the subtleties of the language. During the model training phase, a `WeightedTrainer` is used even after balancing the dataset. This served as an additional layer of equilibrium.

Weighted Cross-Entropy Loss used during training "punished" the model more severely for missing a sample from the minority class if the distribution changed slightly during the 80/20 train-test split, guaranteeing that the model remained equally sensitive to both bullish and bearish signals. To guarantee the repeatability of the results, random `seed=42` was set to govern the creation of the test dataset.

An objective assessment of the model's performance on "live" data replicated a real-world deployment scenario, where the system is forced to understand fresh daily posts by isolating 20% of the testing set.

3.2.5 Probabilistic Inference (Softmax Logic) and Metadata Re-linking

The inference pipeline's execution on the hold-out test set initiates the shift from a trained model to financial intelligence. The model's final layer output (logits) is transformed into a continuous probability distribution (`pred_prob`) between 0 and 1 using Softmax Logic instead of using raw classification labels. This likelihood serves as a "Sentiment Intensity" indicator, with extreme bullishness represented by a score of 0.9 and severe bearishness by a score of 0.1. The market's sentiment is preserved by exporting probabilities rather than binary labels, enabling more accurate econometric modeling.

Metadata Re-linking was carried out after inference. The `created_datetime` and the extracted tickers list are the two original Reddit features that are re-aligned with the synthetic sentiment scores in this stage. This produces a high-fidelity dataset in which every sentiment score is linked to a particular financial item and a particular time.

3.2.6 Multi-Level Aggregation and Ticker-Specific Grouping

The `aggregate_daily_sentiment` module, which converts detailed post-level data into macro-level market signals, is the central component of our analysis. The system does a Ticker-Specific Grouping for each distinct date. For example, the system determines a Weighted Average Sentiment for a stock if "NVDA" appears in fifty different posts on a given Tuesday. In order to reveal the collective "knowledge of the community" about a certain S&P 500 constituent, this aggregation successfully filters out individual outliers by giving high-volume discussion days greater statistical weight than days with isolated mentions.

3.2.7 Time-Series Smoothing and Momentum Signal Generation

Time-Series Momentum Strategy is used to turn fluctuating daily sentiment ratings into useful information. Since raw sentiment is inherently noisy, two main Moving Averages (MA) are computed: a 30-Day MA to set a baseline for long-term institutional sentiment and a 7-Day MA to capture short-term. Sentiment Momentum Oscillator is computed as the difference between these two averages (MA7–MA30) is the final result. In comparison

to the long-term trend, short-term social buzz is accelerating when the momentum value is positive. This crossover provides a complex connection between social discourse and market dynamics and frequently acts as a leading indication for stock breakouts or changes in trading volume in a financial setting.

3.2.8 Sentiment Evaluation and Discussion

With a best F1-Score of 0.9174, the implementation of this NLP pipeline produced a strong discriminative performance, its high accuracy proves sentiment signals dependable for financial modeling. Extreme market stress frequently led to "Panic-Volume Clusters" which shows a non-linear link between post volume and sentiment polarity.

3.2.9 Real-Time Data Retrieval and Temporal Extension

Live Social Media Retrieval is implemented to predict the sentiment in the current market environment. It uses live JSON streams to retrieve the most recent 100 rows of data per subreddit. Multi-Subreddit Broadening Strategy was used to get around the inherent data sparsity of individual forum threads, boosting the sample size to almost 2,000 live data rows per execution, by broadening the retrieval scope over 22 high-impact financial subreddits (i.e. r/wallstreetbets). This extension guarantees that the S&P 500 sentiment signal reflects a diverse cross-section of retail and quantitative discourse rather than being distorted by the peculiar prejudices of a single community.

The project offers a multifaceted view of market health by combining these three different streams: Retail Sentiment (Social), Market Performance (S&P 500), and Institutional Indicators (Bloomberg). By bridging the gap between qualitative social and institutional data, this integrated data structure enables the final model to account for the behavioral causes driving market swings. The final predictive model is able to predict market downturns and trend reversals within a 30-day horizon by integrating Bloomberg's price and volatility indicators with the most recent Reddit posts. This provides a "nowcasted" view of market health that is not possible with historical datasets alone.

4. Problem and Algorithm/Model:

4.1 Problem Definition

Our project addresses an early-warning classification problem for the U.S. equity market: predicting whether the S&P 500 index will enter a downturn over the next 30 trading days.

The practical motivation is asymmetric risk. In crash detection, missing a genuine downturn is typically more costly than issuing a false alert, therefore the modelling objective is not generic accuracy maximization, instead a timely identification of elevated downside risk. This framing is consistent with the broader literature on crash-risk prediction, where machine learning methods are often used to capture nonlinear relationships among market, macroeconomic, and sentiment predictors that simpler linear benchmarks struggle to model robustly. [H Dichtl, 2023]

4.2 Algorithm/Model Initialization

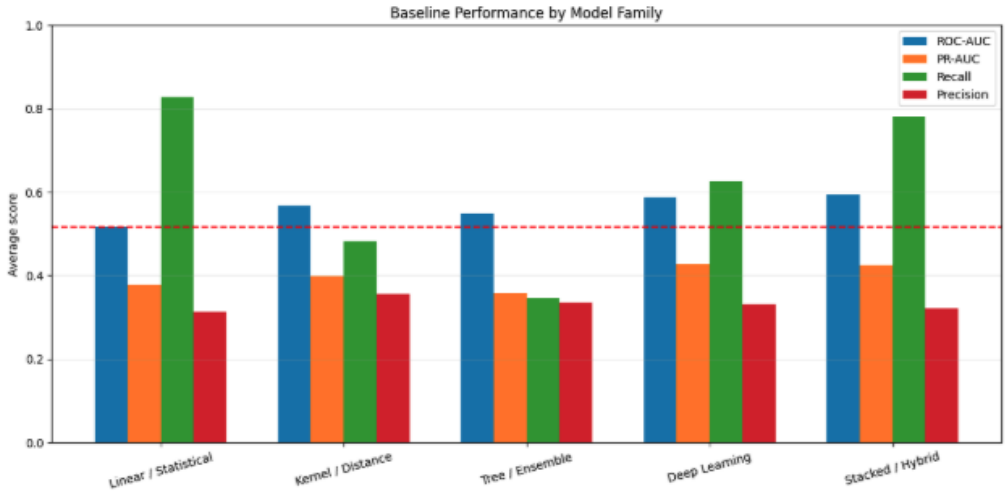


Figure 1: Family level baseline model comparison

The first modelling stage evaluates different model families (models' details in Appendix) with the default settings to determine which architectures exhibit meaningful ranking ability before any expensive tuning is implemented.

Table 4: Model Family

Model Family	**Model**
Linear/Statistical	Logistic Regression, Linear Discriminant Analysis, Quadratic Discriminant Analysis, Gaussian Naive Bayes
Kernel/Distance	Support Vector Machine, K-Nearest Neighbors
Tree/Ensemble	Random Forest, Extra Trees, AdaBoost, Histogram Gradient Boosting, LightGBM, XGBoost, CatBoost
Deep Learning	MLP Neural Net, TabNet
Stacked/Hybrid	Soft Voting Ensemble

The results show a clear contrast between linear or statistical models and non-linear learners. Linear/Statistical models sit the lowest ROC-AUC score among families, producing weak class separation. Meanwhile, non-linear models such as Random Forest, SVM, MLP, and the Voting Ensemble deliver stronger ROC-AUC performance and therefore more credible evidence of useful signal. The objective of this stage is not to optimize every model exhaustively, but rather filtering out models with meaningful ranking ability, and concentrating tuning effort on them. In this sense, the pure financial stage serves as an initial benchmark and a model-screening filter, indicating that the feature set contains predictive information but also showing a practical precision ceiling.

4.3 Modelling Architecture

The final architecture is a stacked multimodal classifier.

At the base level, the system uses three tuned financial learners: Random Forest, Support Vector Machine, and Soft Voting Ensemble. These models are trained on engineered financial features and each of them output a probability of future downturn.

At the meta level, a logistic-regression meta-learner combines those base-model probabilities with three sentiment features (`momentum`, `weighted_sentiment`, and `total_mentions`), plus one interaction term `Financial_Panic_Interaction`, to produce the final risk score.

Through the stacking method applied, we tend to improve generalization by learning how to combine the strengths and biases of multiple lower-level models through out-of-fold predictions rather than naive averaging. [DH Wolpert, 1992 — KM Ting]

The first modelling stage establishes what can be achieved using financial features alone. This stage is significant since it provides the benchmark performance of a pure market-data approach and keeps the strongest models. The second modelling stage then identifies whether the integration of sentiment data adds meaningful information that can improve the precision-recall trade-off in an early-warning setting.

Recent studies show that machine learning models can outperform traditional crash-prediction baselines when the relationship between predictors and crash events is nonlinear and time-varying. Related work also found that multimodal inputs, including sentiment-derived text signals, can improve crash or distress prediction beyond the market data alone. Therefore, our project setup structures a transparent stacked design that combines stable financial models with a sentiment-enhanced meta-layer, rather than relying on a single monolithic model. [H Dicthl, 2023 — Y Sheng, 2024]

5. Evaluation Methodology:

5.1 Data Usage & Chronological Splitting Design

The modelling pipeline is chronological and leakage-aware throughout. In the first half, the financial dataset is split into train, validation, and fully unseen test periods in time order. In the second half, the financial data is restricted to the 2008–2025 window so that it matches the Reddit sentiment coverage window. The combined dataset is then split into 2008–2021 for meta-training and 2022–2025 for an unseen blind test. After the architecture is fixed, the final production models are retrained on 2008–2021, while Jan–Apr 2026 is kept fully outside training and used for live simulation.

5.2 TimeSeriesSplit & OOF bridge generation

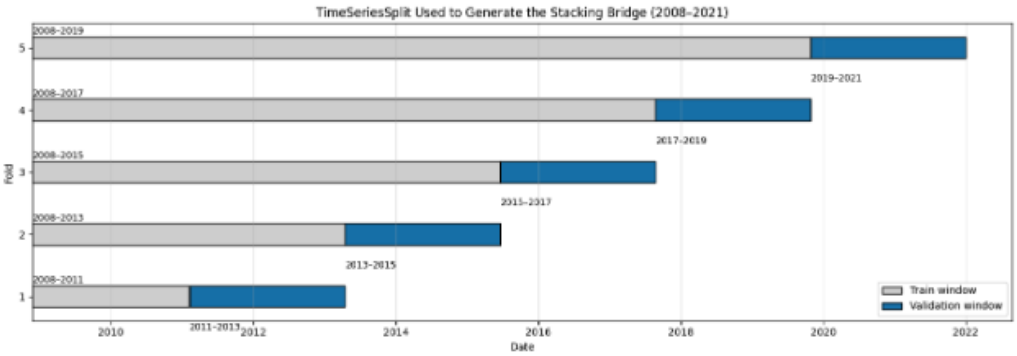


Figure 2: TimeSeriesSplit visualization

The second stage constructs the stacking bridge using out-of-fold (OOF) predictions. For the 2008–2021 training period, `TimeSeriesSplit` is applied to generate OOF probabilities from the three retained base learners. As shown in Figure 3, each fold uses an earlier block for training and a later block for validation, with the validation windows moving forward through time. The meta learner is not trained on in-sample base probabilities, but on out-of-fold probabilities generated from models that did not see those validation rows during fitting. Once the bridge is built for the training period, the base models were retrained on the full pre-2022 window and used to generate level-1 probabilities for the unseen 2022–2025 test period.

5.3 Sentiment Alignment, Aggregation and Missing-value Handling

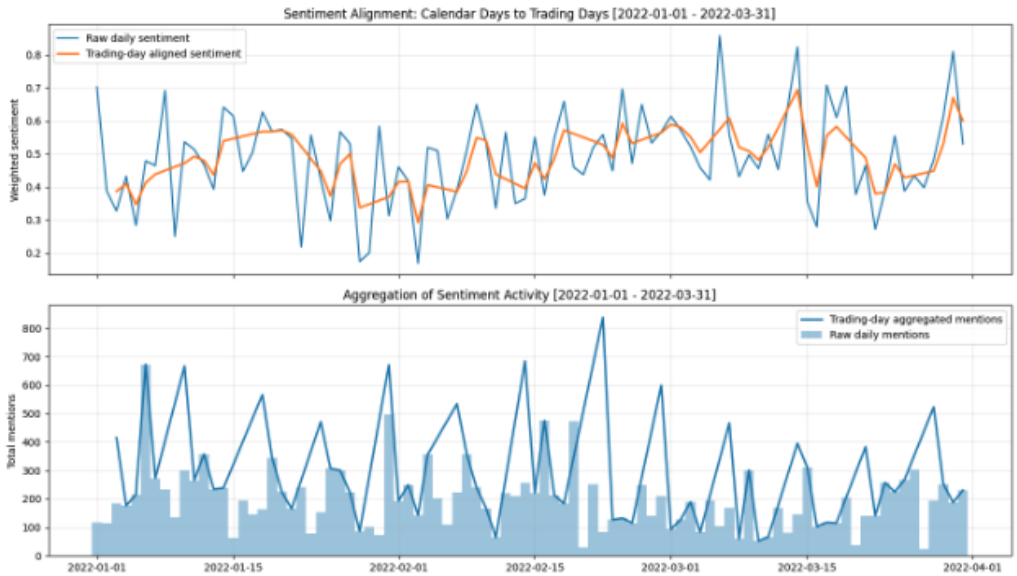


Figure 3: Sentiment alignment & calendar-trading-days aggregation

Sentiment integration is handled through a calendar-to-market alignment procedure. The Reddit dataset is first smoothed using Exponential Moving Averages (EMA) for `sentiment` and `momentum`. To illustrate better, referring to the top panel of Figure 4, which compares

the raw daily sentiment with the trading-day aligned sentiment series, we can observe the aligned series follow the same medium-term rises and falls as the raw series, with smaller day-to-day swing. Therefore, high-frequency noise is reduced through EMA and trading-day mapping.

`total_mentions` is also mapped from calendar days to trading days using forward alignment, such that weekend and holiday sentiment are carried into the next relevant market session rather than discarded. As shown in the bottom panel, the peaks in the aggregated line are often larger than individual raw-day bars because multiple calendar-day observations, including weekends are rolled into the next trading session.

For the missing values, `momentum` and `total_mentions` are filled with zero, and `weighted_sentiment` is filled with 0.5 to represent a neutral fallback.

5.4 Evaluation Metrics and Thresholding Strategy

The evaluation metrics are chosen to match the project's asymmetric objective. ROC-AUC and PR-AUC are used to evaluate a model's ranking ability before a decision threshold is chosen. Recall is the primary operating metric throughout the modelling pipeline. Precision remains important since a model that issues too many false alarms becomes unusable even if the recall is high. To balance these objectives, we apply the F2 score to weight recall more heavily than precision. The first half hyperparameter tuning and threshold analysis therefore emphasize whether the models can retain high recall without collapsing precision.

The thresholding procedure is split into two parts. In the validated binary classifier, the meta learner threshold is selected on the training side and then locked before testing. Within the 2008–2021 meta-training period, the resulting fold-wise optimal thresholds are averaged, producing a locked threshold of 0.28.

A higher 0.35 threshold is introduced later, which is not part of formal model validation. It is a manual communication threshold used to separate "Elevated Risk" from "Crash Warning" in the 2026 visualization.

6. Results and Discussion:

6.1 Results

6.1.1 Financial-only Results & Precision Ceiling

The first half results show that the financial dataset contains meaningful predictive information, but the signal is not strong enough to support a practical high-recall warning system.

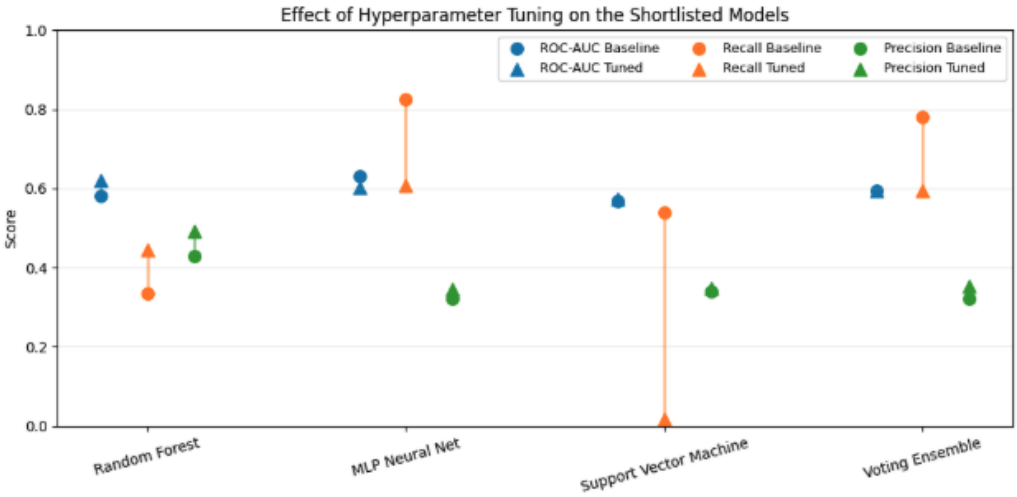


Figure 4.1: Hyperparameter Tuning Summary

With the hyperparameter tuning, the retained non-linear models move into a narrower performance band, with ROC-AUC values clustered roughly between 0.58–0.62. The result from threshold tuning stage shows that precision plateaus at a relatively low level when recall changes much more sharply. The recall differences come from how each tuned model converts probabilities into class labels at the default threshold, not only from how well the model ranks observations. For example, SVM still has a reasonable ROC-AUC while producing a recall of nearly zero since its tuned version became more conservative and failed to catch the downturn cases. Therefore, this implies that hyperparameter tuning does not affect all metrics in the same way, where ROC-AUC remains relatively stable, but recall is highly sensitive to how each model's tuned probabilities distribution interacts with the default decision threshold.

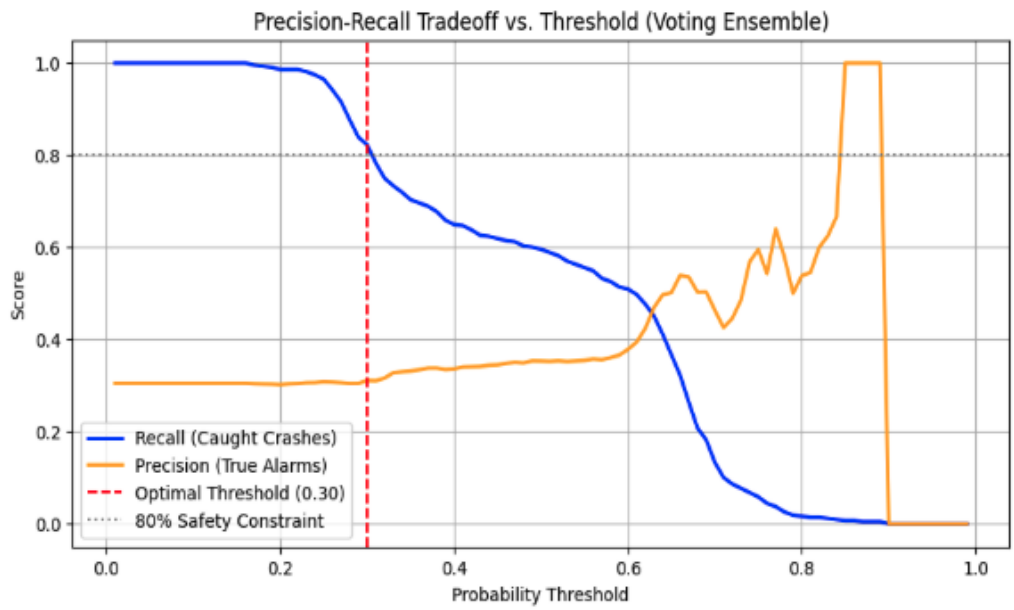


Figure 4.2: Threshold Tuning Summary

After hyperparameter tuning, threshold tuning provides another insight. When recall is

constrained near 0.8 (desired region), precision improves only up to a limited point (∼0.3). Therefore, as a conclusion of the first half modelling stage, the pure financial dataset is only able to detect market fragility, but not at a precision level that minimizes the rate of false alarms.

6.1.2 Final Stacked-Model Performance on the Blind 2022-2025 Test

With the sentiment features aligned and combined with the retained models, the final meta learner produces a better blind-test operating point. In the unseen 2022–2025 period, the locked threshold of 0.28 yields recall of approximately 0.9676 and precision of approximately 0.4176. Relative to the earlier financial-only modelling stage, the stacked model retains high recall while moving precision upward by approximately 12%.

6.1.3 Interpretation of the Meta Learner

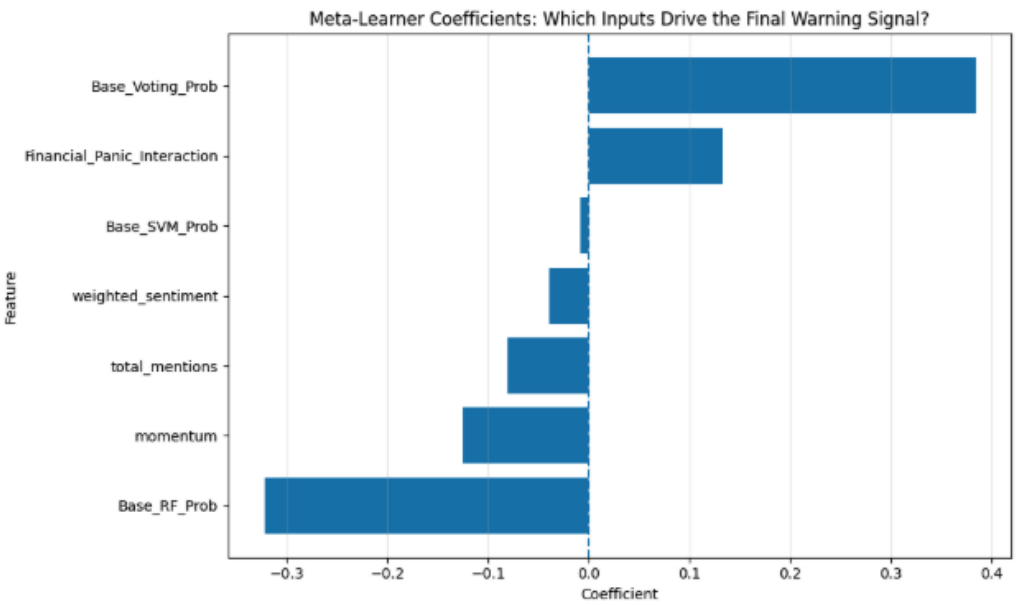


Figure 5: Meta learner coefficient plots

Within the meta learner, the coefficient structure provides useful insight into how the stacked model works. `Base_Voting_Prob` is the strongest positive contributor, while `Financial_Panic_Interaction` is the second strongest positive contributor. This indicates that the final signal is driven not only by the financial ensemble's market-fragility estimate, but also by situations where the financial stress is reinforced by sentiment momentum. By contrast, `Base_RF_Prob`, `momentum`, `total_mentions`, `weighted_sentiment` and `Base_SVM_Prob` carry negative coefficients in the final meta learner. This indicates the meta learner learns a structured weighting rule in which sentiment sometimes moderates, rather than amplifies raw financial alarm signals.

6.1.4 Predictions for Jan - Apr 2026

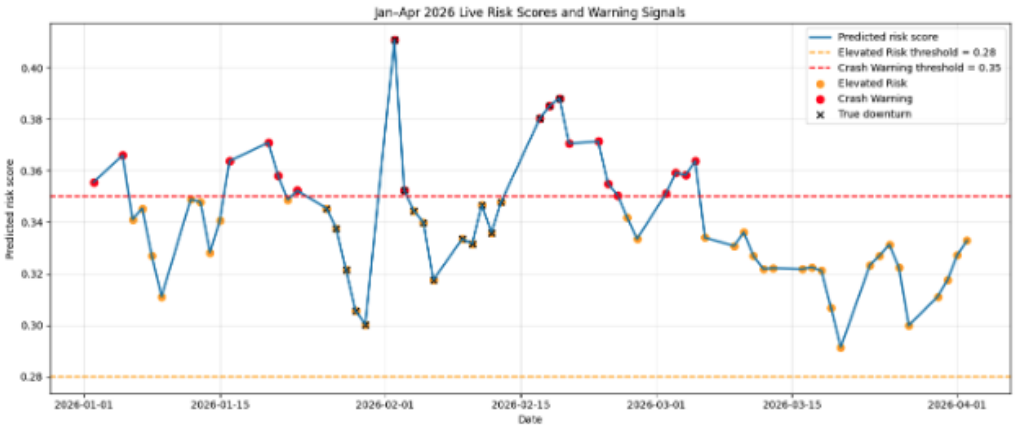


Figure 6: Final prediction trend line

The binary operating threshold remains at 0.28 since it comes directly from the validated meta learner procedure and delivers the blind-test operating point the project seeks to preserve. The additional 0.35 threshold is introduced in the Jan–Apr 2026 live simulation, creating three risk bands which are an interpretation layer on top of the predicted risk score.

Normal means the score is below 0.28, Elevated Risk means the score is between 0.28–0.35, and Crash Warning means the score is at least 0.35. In practical terms, a day in the Elevated Risk or Crash Warning band means the model is indicating a higher likelihood that a downturn case may occur within the next 30 trading days, with Crash Warning representing the stronger signal.

This additional banding improves readability because the validated locked threshold is relatively low at 0.28. If we only used the binary rule, then almost every score above 0.28 will be detected as downturn even though the score rises much further from the threshold. The introduction of 0.35 separates moderate warning states from stronger warning states instead of collapsing them into one class.

Since the true `downturn_target` labels stop at 19-02-2026, only the start of January to mid-February 2026 can be compared directly against observed outcomes. From the plot, the model successfully identified all actual downturn cases. The model also achieves a precision of 54.5% in the window, implying 54.5% of the days flagged as positive were truly followed by a downturn case. The model is catching everything, but it is also warning more often than necessary. A 0.857 F2 score indicates the model performs well under a recall-first objective, preserving full downside coverage while maintaining moderate precision.

6.2 GUI

6.2.1 Public Dashboard: S&P 500 Downturn Visualisation

6.2.1.1 Overview

The public dashboard provides an interactive visualisation of the ensemble model's downturn predictions against observed S&P 500 price movements. For a selected prediction

date, the dashboard plots the closing price of the S&P 500 index over the subsequent 30 trading days and evaluates whether the model's binary prediction was correct.

6.2.1.2 Data Sources

S&P 500 Index Prices:

Historical daily price data was sourced directly from the raw dataset from Bloomberg Terminal (`SPX Last Open val.xlsx`). The closing price field, denoted `PX_LAST` in Bloomberg's data convention, is used exclusively as the reference price throughout all calculations.

Model Predictions:

The model's output is stored in `jan_apr_2026_predictions.csv`, a precomputed file generated by the ensemble ML pipeline. This file covers trading days from 2 January 2026 to 2 April 2026 and contains the following columns relevant to the dashboard:

`Predicted_Risk_Score`	The ensemble model's continuous risk estimate, ranging from 0 (low risk) to 1 (high risk).
`Crash_Warning_Flag`	A binary prediction derived from the risk score. A value of 1 is assigned when `Predicted_Risk_Score` > 0.28, indicating the model predicts a potential downturn.
`True_Downturn`	The ground truth label. A value of 1 indicates that the S&P 500 index experienced a closing price decline of at least 3% relative to the selected date's closing price at some point within the subsequent 30 trading days.

6.2.1.3 Methodology

Upon selecting a prediction date, t, the closing price on that date, P_0, is extracted from the Bloomberg dataset and used as the reference price. The 3% downturn threshold is defined as:

$$P_{threshold} = P_0 \times 0.97$$

A breach is recorded on any subsequent trading day $t + k$ where the closing price P_{t+k} satisfies:

$$P_{t+k} < P_{threshold}$$

All such breach days within the 30-trading-day window are identified and marked on the chart.

The visualisation displays S&P 500 closing prices from the selected date through the next 30 trading days. Calendar weekends and market holidays are excluded, as the dataset contains only trading day observations. Where the dataset ends before a full 30-trading-day window is available, the chart displays only the available days and indicates the partial window to the viewer.

A verified cutoff date is dynamically computed as the latest prediction date for which a minimum of 30 trading days of S&P 500 price data exists beyond it in the dataset. For the current dataset ending April 2, 2026, this cutoff falls on approximately February 19, 2026. Prediction dates on or before the cutoff are considered fully verified, meaning the actual outcome over the complete 30-day window can be determined. Prediction dates beyond the cutoff are classified as pending, as the full outcome window has not yet elapsed.

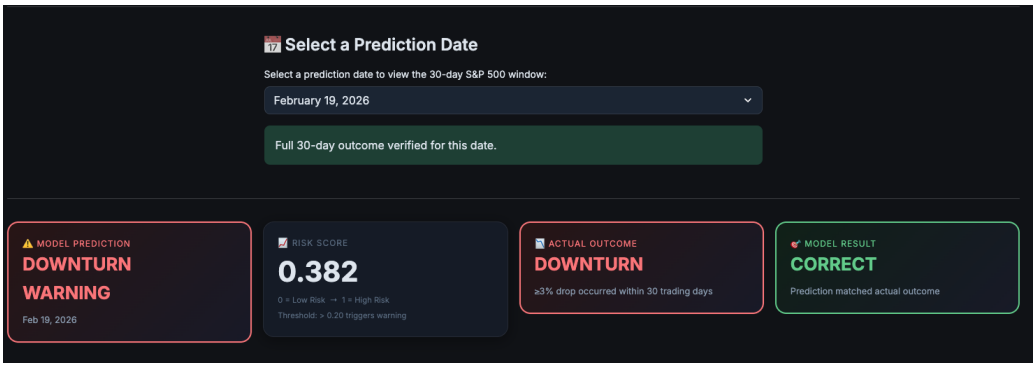
6.2.1.4 Prediction Outcome Evaluation

For verified dates, the model's binary prediction (`Crash_Warning_Flag`) is compared against the ground truth label (`True_Downturn`) to produce one of four outcomes:

Table 5: Prediction Results

Model Prediction	Actual Outcome	Verdict
Downturn (flag = 1)	Downturn (label = 1)	Correct
Downturn (flag = 1)	No downturn (label = 0)	False Alarm
Safe (flag = 0)	Downturn (label = 1)	Missed
Safe (flag = 0)	No downturn (label = 0)	Correct

For unverified dates, the outcome is displayed as Pending Verification, as the 30-trading-day window extends beyond the available price data. The verdict is surfaced to the user through a colour-coded banner above the chart: green for a correct prediction, red for an incorrect prediction, and amber for a pending outcome.



6.2.1.5 Sentiment & Momentum Analysis

The sentiment engine collects texts daily and computes a weighted average score. The score of 0.50 is considered the "Neutral Pivot".

- Score > 0.5 indicates "Bullish Bias": more positive news.
- Score < 0.50 indicates "Bearish Bias": more negative news.

Momentum (M) is calculated as the percentage difference between the current price and the price n days ago:

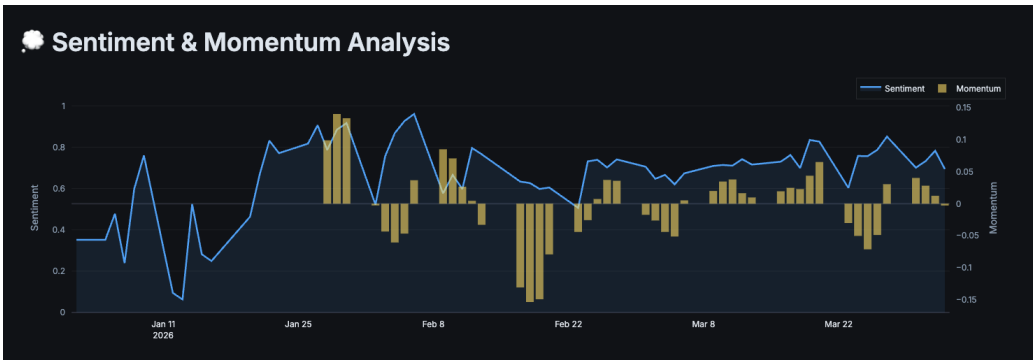
$$M = \frac{P_{Current} - P_n}{P_n}$$

The gold bars in the graph represent this speed on the secondary Y-axis (-0.15 to $+0.15$).

- For Expansion: The bars becoming larger represent a more powerful upward trend.
- For Contraction: The bars becoming smaller represent a more powerful downward trend.

Both Sentiment and Momentum are mapped against each other via the timeline X-axis, allowing user to detect Convergence and Divergence:

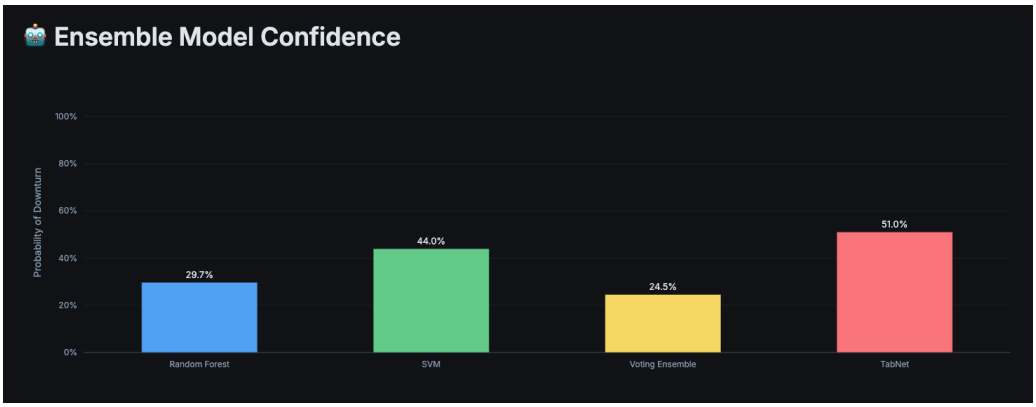
- Convergence (Confirmation): Sentiment and momentum moving in same direction (e.g. both rising – signals a strong established trend).
- Divergence (Warning): Opposing movements (e.g. high sentiment paired with falling momentum), acts as a leading indicator for trend reveals.



6.2.1.6 Ensemble Model Confidence

The bar chart illustrates the probability of a downturn in four separate models on a particular day (between January to March 2026):

- Individual Models: Includes risk probabilities from Random Forest (29.7%), SVM (44.0%), and TabNet (51.0%).
- Voting Ensemble (24.5%): This is the overall prediction produced by the model.
- Threshold Logic: As the Ensemble probability (24.5%) is greater than the threshold value (20%), a Downturn Alert is activated by the system.



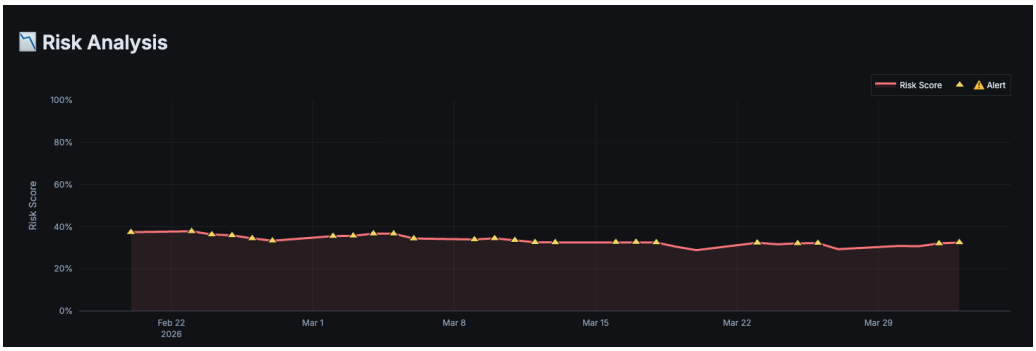
6.2.2 Admin Dashboard: Model Performance and Social Analytics

6.2.1 Overview

The Admin Dashboard offers an overall perspective of the backend prediction algorithm. Its emphasis lies on risk sustainability, social media sentiment, and distribution statistics, thereby ensuring that the model is both reliable and logical.

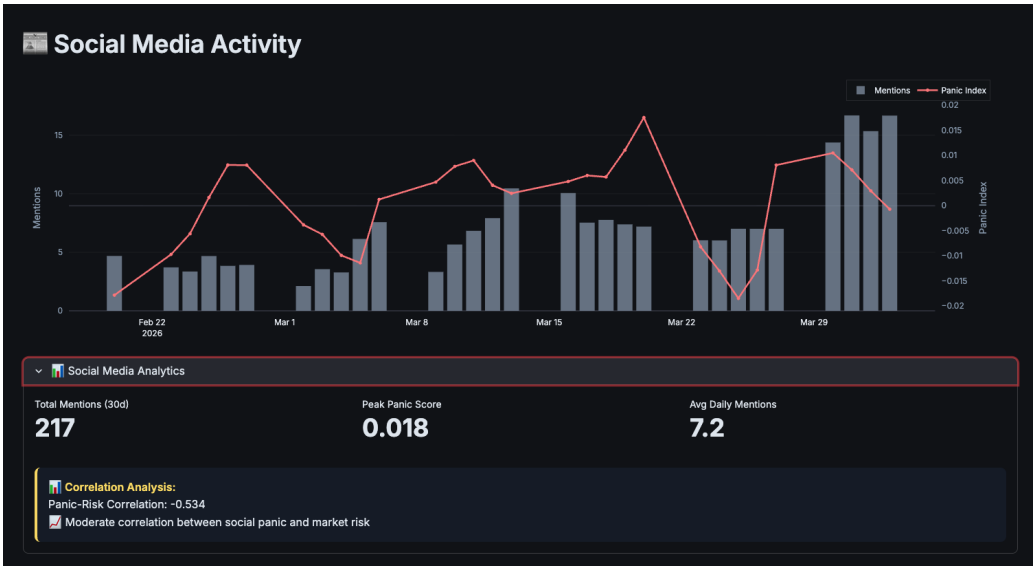
6.2.2 Risk Analysis

Risk Level Tracking: The daily risk levels depict the consistent high-risk environment (30% – 40%) experienced during late February and March 2026.



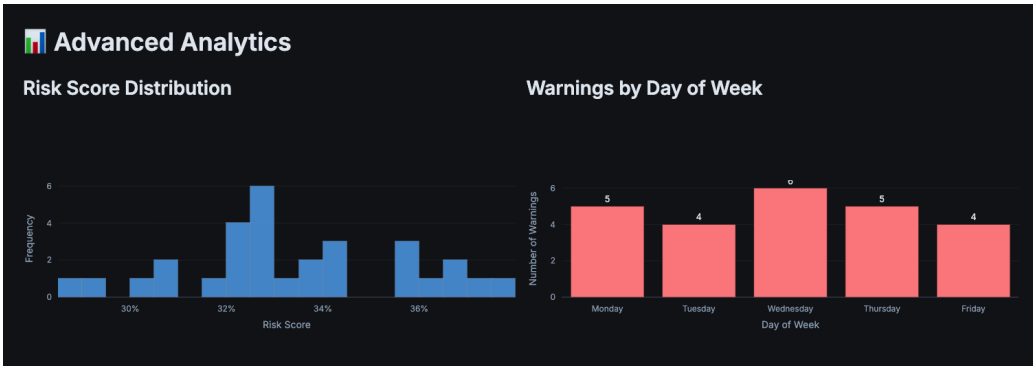
6.2.3 Social Media Analysis

Panic vs Activity: Correlation analysis between volume (mentions) and the "Panic Index" indicates the degree of market panic.



6.2.4 Advanced Analytics

This histogram illustrates the risk distribution, which peaks at 33%. The other histogram shows the days of the week when warnings occur, revealing the most active days (e.g. Wednesday for January–April 2026) and help detect cyclical patterns.



7. Appendix

7.1 Datasets

Table 2: Bloomberg Engineered Features

Table 2: Bloomberg Engineered Features

#	Feature Name	Computation / Formula	Rationale
A. SPX Price & Return Features			
1	SPX_PX_LAST	Raw closing price	Anchor price for all return and distance calculations.
2	SPX_PX_VOLUME	Raw daily traded volume	Volume confirms price moves; abnormal volume during declines signals institutional selling.
3	SPX_log_ret_1d	$\ln(P_t/P_{t-1})$	Daily momentum; captures immediate price pressure. Log-returns preferred for statistical properties.
4	SPX_log_ret_5d	$\ln(P_t/P_{t-5})$	1-week momentum; identifies short-term trend direction.
5	SPX_log_ret_21d	$\ln(P_t/P_{t-21})$	1-month momentum; standard feature in all ML crash-prediction models.
6	SPX_log_ret_63d	$\ln(P_t/P_{t-63})$	1-quarter momentum; captures medium-term trend regime.
7	SPX_log_ret_126d	$\ln(P_t/P_{t-126})$	6-month momentum; identifies longer-term bull/bear regime.
B. SPX Volatility Features			
8	SPX_GK_vol_5d	Garman-Klass estimator, 5-day rolling window	OHLC-based volatility is up to 8x more efficient than close-to-close. 5-day window captures immediate volatility spike.
9	SPX_GK_vol_10d	Garman-Klass estimator, 10-day rolling window	2-week volatility; bridges daily spikes and monthly baseline.
10	SPX_GK_vol_21d	Garman-Klass estimator, 21-day rolling window	1-month realized volatility; standard risk measure for institutional portfolios.
11	SPX_GK_vol_63d	Garman-Klass estimator, 63-day rolling window	Quarterly volatility baseline; used as denominator in regime-change ratios.
12	SPX_vol_ratio_5_63	GK_vol_5d / GK_vol_63d	Short-to-long vol ratio. Ratio > 1.5 signals a volatility regime transition — a leading precursor to drawdowns.
13	SPX_vol_ratio_21_63	GK_vol_21d / GK_vol_63d	Monthly-to-quarterly vol ratio. Identifies sustained volatility elevation vs. a single-day spike.
C. SPX Sharpe Ratios & Drawdown from Peak			
14	SPX_sharpe_21d	Ann. mean return / Ann. std dev, 21-day window	Deteriorating Sharpe signals risk-return breakdown. Identified as top SHAP feature in Benhamou & Ohana (2021) crash model.

Table 2: Bloomberg Engineered Features (continued)

#	Feature Name	Computation / Formula	Rationale
15	SPX_sharpe_63d	Ann. mean return / Ann. std dev, 63-day window	Quarterly risk-adjusted return; captures medium-term deterioration before price fully confirms it.
16	SPX_sharpe_126d	Ann. mean return / Ann. std dev, 126-day window	6-month Sharpe; identifies prolonged regime deterioration similar to the pre-crash pattern found in academic literature.
17	SPX_drawdown_21d	$(P_t/\max(P_{t-21:t})) - 1$	Proximity to 1-month peak. Index already off its high is more likely to continue declining (momentum continuation).
18	SPX_drawdown_63d	$(P_t/\max(P_{t-63:t})) - 1$	Proximity to quarterly peak. Distance-from-high features account for up to 36% of SHAP importance in published models.
19	SPX_drawdown_126d	$(P_t/\max(P_{t-126:t})) - 1$	6-month drawdown depth; captures bear market context beyond short-term corrections.
20	SPX_drawdown_252d	$(P_t/\max(P_{t-252:t})) - 1$	Annual drawdown; identifies whether the index is in a sustained peak-to-trough decline vs. a minor pullback.
D. SPX Moving Average Features			
21	SPX_vs_MA_50	$(P_t/\text{SMA}_{50}) - 1$	Continuous distance from 50-day MA; captures short-term trend strength without the lag of a binary crossover signal.
22	SPX_vs_MA_200	$(P_t/\text{SMA}_{200}) - 1$	Distance from 200-day MA; price below this level is a widely confirmed bearish regime indicator.
23	SPX_below_MA200	1 if $P_t < \text{SMA}_{200}$, else 0	Binary regime flag. Captures the non-linear threshold effect at the 200-day MA that continuous distance may underweight.
24	SPX_MA50_slope	pct_change(SMA_50, periods=10)	10-day slope of 50-day MA; shows whether the short-term trend is accelerating or rolling over.
25	SPX_MA200_slope	pct_change(SMA_200, periods=10)	10-day slope of 200-day MA; a flattening or declining long-term MA slope confirms a bearish regime shift.
E. Higher Moments, Intraday Structure & Volume			
26	SPX_skew_21d	Rolling 21-day skewness of daily log-returns	Negative skewness indicates a left-tailed return distribution — a direct statistical precursor to crash risk.
27	SPX_kurt_21d	Rolling 21-day kurtosis of daily log-returns	Elevated kurtosis signals fat tails and elevated probability of extreme negative outcomes.
28	SPX_autocorr_21d	Rolling 21-day lag-1 autocorrelation of returns	Positive autocorrelation indicates trending/crisis regime; negative indicates mean-reversion. Changes in sign precede regime transitions.

Table 2: Bloomberg Engineered Features (continued)

#	Feature Name	Computation / Formula	Rationale
29	SPX_intraday_ret	(Close − Open)/Open	Captures within-day directional pressure. Persistent negative intraday returns signal institutional distribution.
30	SPX_range_pct	(High − Low)/Open	Intraday volatility proxy. Widening range during declines is characteristic of high-stress sessions.
31	SPX_volume_chg_1d	pct_change(Volume, periods=1)	Day-over-day volume surge alongside declining prices ('distribution days') signals institutional selling.
32	SPX_volume_MA_20	Rolling 20-day mean of Volume	Baseline volume level used to contextualise current trading activity.
33	SPX_volume_vs_MA20	$(\mathrm{Vol}/\mathrm{Vol}_{\mathrm{MA20}}) - 1$	Volume deviation from norm. Abnormally high volume on down days confirms panic or forced selling.
34	SPX_volume_zscore_63	$(Volume - \mathrm{mean}_{63})/\mathrm{std}_{63}$	Rolling z-score normalises volume across different market regimes, allowing scale-invariant comparisons.
F. Technical Indicator Features			
35	SPX_RSI_2	RSI with 2-day lookback	Very short-term RSI captures extreme oversold/overbought conditions with contrarian signal properties at 1–5 day horizons.
36	SPX_RSI_5	RSI with 5-day lookback	1-week RSI; bridges ultra-short RSI(2) and the standard RSI(14), capturing short-term momentum deterioration.
37	SPX_RSI_14	RSI with 14-day lookback	Standard RSI; persistent readings below 50 confirm downtrend; failure to recover above 50 signals sustained weakness.
38	SPX_BB_width	$(2 \times \mathrm{std}_{20})/\mathrm{SMA}_{20}$	Bollinger Band squeeze (low width) often precedes sharp directional moves. Expansion signals a volatility regime change.
39	SPX_BB_pct_b	(Close − LB)/(UB − LB)	Position within the Bollinger Band. Sustained readings below 0.2 confirm the index is in a downtrend.
40	SPX_MACD_hist	SMA(12) − SMA(26) − SMA(MACD, 9) [SMA used to avoid EMA leakage]	MACD histogram turning negative signals momentum deterioration before price confirms it, providing an early warning.
41	SPX_MACD_hist_chg	First difference of MACD histogram	Rate of change of momentum — an even earlier signal than the histogram level alone.
G. VIX Features			
42	VIX_PX_HIGH	Raw daily VIX high (used as level proxy)	VIX is the most predictive single feature family for 30-day equity stress across nearly all published studies. Levels above 20, 25, and 30 mark distinct risk regimes.

Table 2: Bloomberg Engineered Features (continued)

#	Feature Name	Computation / Formula	Rationale
43	VIX_chg_1d	$\text{VIX}_t - \text{VIX}_{t-1}$	Absolute daily change; captures sudden fear spikes that percentage change may understate at low VIX levels.
44	VIX_ret_1d	pct_change(VIX, periods=1)	Daily VIX momentum; lagged VIX returns showed highest feature importance in Random Forest and XGBoost crisis models.
45	VIX_ret_5d	pct_change(VIX, periods=5)	A 5-day VIX increase exceeding 30–40% is identified in literature as a powerful near-term drawdown signal.
46	VIX_ret_10d	pct_change(VIX, periods=10)	2-week VIX momentum; captures sustained fear escalation vs. a single-day spike reverting quickly.
47	VIX_range	VIX_PX_HIGH − VIX_PX_LOW	Novel 'ULD' feature. Intraday VIX range proxies uncertainty about the level of fear itself. Literature reports out-of-sample $R^2 \approx 4\%$, exceeding traditional predictors.
48	VIX_range_5d_avg	Rolling 5-day mean of VIX_range	Persistent VIX range expansion is more indicative of a stressed regime than a single-day VIX candle.
49	VIX_range_21d_avg	Rolling 21-day mean of VIX_range	Monthly average of daily VIX range; captures whether volatility-of-volatility is chronically elevated.
50	VIX_MA_20	Rolling 20-day SMA of VIX	Short-term VIX trend baseline; used in VIX_vs_MA_20 to measure current fear vs. recent norm.
51	VIX_MA_60	Rolling 60-day SMA of VIX	Longer-term VIX baseline; denominator for term structure proxy.
52	VIX_term_struct_proxy	VIX / VIX_MA_60	Approximates VIX futures backwardation without futures data. Ratio > 1.2 signals near-term hedging demand exceeds long-term expectations — elevated crash risk.
53	VIX_vs_MA_20	$(\text{VIX}/\text{VIX}_{\text{MA20}}) - 1$	Short-term VIX deviation from its recent average; captures whether fear has suddenly escalated above the recent baseline.
54	VIX_pct_rank_252	Rolling 252-day percentile rank of VIX	Contextualises absolute VIX level within its annual distribution. A high absolute VIX during a persistently fearful period carries less incremental signal than the same reading after a long calm.
55	VIX_above_20	1 if VIX > 20, else 0	Binary flag for 'elevated risk' regime. VIX > 20 marks the threshold where drawdown probabilities begin to rise non-linearly.
56	VIX_above_25	1 if VIX > 25, else 0	'High fear' regime flag. Captures the intermediate stress zone between elevated risk and extreme panic.

Table 2: Bloomberg Engineered Features (continued)

#	Feature Name	Computation / Formula	Rationale
57	VIX_above_30	1 if VIX > 30, else 0	'Extreme stress' regime flag. Above 30 corresponds to historic crisis episodes with high 30-day drawdown probability.
58	VIX_vol_5d	Rolling 5-day std dev of VIX daily returns	Volatility-of-VIX (vol-of-vol) — measures uncertainty about the level of fear, a characteristic of regime transitions.
59	VIX_vol_21d	Rolling 21-day std dev of VIX daily returns	Monthly vol-of-vol; elevated values indicate prolonged market uncertainty rather than a transient fear spike.
H. Treasury Yield Curve Features			
60	USGG10YR_PX_LAST	Raw 10-year Treasury yield	Benchmark long-term rate; used to construct yield spreads and rate-of-change signals.
61	USGG2YR_PX_LAST	Raw 2-year Treasury yield	Short-term rate reflecting Fed policy expectations; rapid rises signal financial conditions tightening.
62	USGG3M_PX_LAST	Raw 3-month Treasury yield	Ultra-short rate; used in 10Y–3M spread, the most reliable recession-predicting yield curve measure.
63	US10Y_chg_1d	$\text{US10Y}_t - \text{US10Y}_{t-1}$	Daily long-end yield move; sudden rises can signal growth optimism or inflation fears tightening financial conditions.
64	US10Y_chg_10d	$\text{US10Y}_t - \text{US10Y}_{t-10}$	2-week long-end rate change; captures whether a yield move is sustained or a daily noise event.
65	US2Y_chg_1d	$\text{US2Y}_t - \text{US2Y}_{t-1}$	Daily short-end yield change; proxies immediate shifts in Fed rate expectations.
66	US2Y_chg_10d	$\text{US2Y}_t - \text{US2Y}_{t-10}$	Rapid short-end rises over 10 days directly proxy financial conditions tightening.
67	US3M_chg_1d	$\text{US3M}_t - \text{US3M}_{t-1}$	Daily 3-month yield change; most sensitive to imminent Fed policy shifts.
68	YC_10Y_2Y	USGG10YR − USGG2YR	Primary yield curve slope. Has inverted before every U.S. recession since the 1970s; level provides regime context.
69	YC_10Y_3M	USGG10YR − USGG3M	Academically preferred recession predictor. Inversion of this spread has the highest historical precision for recession forecasting.
70	YC_2Y_3M	USGG2YR − USGG3M	Front-end slope; captures the near-term shape of the policy rate expectations curve.
71	YC_10Y_2Y_chg_10d	$s_{\text{10Y2Y},t} - s_{\text{10Y2Y},t-10}$	Rapid steepening after inversion (de-inversion) is the actionable stress signal for the 30-day horizon, coinciding with the onset of economic deterioration.

Table 2: Bloomberg Engineered Features (continued)

#	Feature Name	Computation / Formula	Rationale
72	YC_10Y_2Y_chg_20d	$s_{10Y2Y,t} - s_{10Y2Y,t-20}$	1-month spread change; captures the full de-inversion dynamic over a policy-relevant horizon.
73	YC_10Y_3M_chg_10d	$s_{10Y3M,t} - s_{10Y3M,t-10}$	Rate of change of the most reliable recession-predictor spread; rapid steepening from deep inversion is the danger signal.
74	YC_inverted_10Y2Y	1 if YC_10Y_2Y < 0, else 0	Binary inversion flag; activates the regime in which de-inversion risk becomes relevant.
75	YC_inverted_10Y3M	1 if YC_10Y_3M < 0, else 0	Inversion of the academically preferred spread; historically the strongest recession predictor.
76	YC_deinversion_signal	1 if YC_10Y_2Y ≥ 0 after ≥ 60 days of inversion	Captures the specific de-inversion event identified as the highest-risk transition. Activation typically precedes equity drawdowns by days to weeks.
77	US2Y_vs_MA20	$(\text{US2Y}/\text{SMA}_{20}(\text{US2Y}))-1$	2-year yield deviation from recent norm; measures whether short-end rates have accelerated above their recent trend, signalling tightening.
I. Gold & Oil Features			
78	XAUUSD_PX_LAST	Raw gold spot price (USD/oz)	Anchor for gold return and ratio calculations.
79	CL1_PX_LAST	Raw WTI crude front-month price	Anchor for oil return and ratio calculations; WTI used as primary oil benchmark.
80	GOLD_log_ret_1d	$\ln(\text{Gold}_t/\text{Gold}_{t-1})$	Daily gold return; confirms immediate risk-off sentiment on the same day as equity stress.
81	GOLD_log_ret_5d	$\ln(\text{Gold}_t/\text{Gold}_{t-5})$	1-week gold return; captures short-term safe-haven capital flow.
82	GOLD_log_ret_21d	$\ln(\text{Gold}_t/\text{Gold}_{t-21})$	1-month gold return; used to construct the relative strength safe-haven signal.
83	GOLD_rel_strength_21d	GOLD_log_ret_21d − SPX_log_ret_21d	Directly measures safe-haven capital flight. Positive values indicate investors rotating out of equities into gold — a classic risk-off pattern (Baur & Lucey, 2010).
84	GOLD_SPX_ratio	XAUUSD / SPX_PX_LAST	Level of gold relative to equities; rising ratio over time confirms a sustained risk-off regime.
85	GOLD_vol_21d	Rolling 21-day std dev of gold log-returns $\times\sqrt{252}$	Gold volatility spikes during extreme events; elevated gold vol alongside equity declines confirms systemic stress.

Table 2: Bloomberg Engineered Features (continued)

#	Feature Name	Computation / Formula	Rationale
86	OIL_log_ret_1d	$\ln(\mathrm{WTI}_t/\mathrm{WTI}_{t-1})$ [clipped at \$0.01 floor]	Daily oil momentum; captures immediate supply/demand shocks. Floor prevents undefined log values from the April 2020 negative WTI event.
87	OIL_log_ret_21d	$\ln(\mathrm{WTI}_t/\mathrm{WTI}_{t-21})$	1-month oil return; large spikes have preceded 4 major recessions where SPX declined 20–48% peak-to-trough.
88	OIL_vol_21d	Rolling 21-day std dev of oil log-returns $\times\sqrt{252}$	Oil volatility; sustained elevation indicates ongoing supply uncertainty rather than a resolved one-day shock.
89	OIL_WTI_Brent_spread	CL1_PX_LAST − CO1_PX_LAST	WTI–Brent spread; a more negative spread signals global supply disruption (Brent premium rises) rather than U.S.-specific demand, a more equity-bearish signal.
90	OIL_WTI_Brent_spread_chg	5-day change in WTI–Brent spread	Rate of change of the spread; rapid widening confirms an accelerating geopolitical or supply shock.
91	GOLD_OIL_ratio	XAUUSD / CL1_PX_LAST	Rising gold-to-oil ratio signals that policy/credit risk is dominating over economic growth as the market driver — a characteristic of recession-risk environments.
J. Cross-Index Breadth & Divergence Features			
92	DJ_log_ret_1d	$\ln(\mathrm{DJIA}_t/\mathrm{DJIA}_{t-1})$	Daily Dow Jones return; cross-index return divergence from SPX on the same day flags breadth deterioration.
93	NDX_log_ret_1d	$\ln(\mathrm{NDX}_t/\mathrm{NDX}_{t-1})$	Daily Nasdaq return; Nasdaq vs. SPX divergence is identified as a key crash predictor in Benhamou & Ohana (2021).
94	MXWO_log_ret_1d	$\ln(\mathrm{MXWO}_t/\mathrm{MXWO}_{t-1})$	Daily MSCI World return; divergence from SPX distinguishes US-idiosyncratic risk from globally synchronised stress.
95	DJ_log_ret_21d	$\ln(\mathrm{DJIA}_t/\mathrm{DJIA}_{t-21})$	1-month DJIA return; used in cross-index dispersion calculation.
96	NDX_log_ret_21d	$\ln(\mathrm{NDX}_t/\mathrm{NDX}_{t-21})$	1-month Nasdaq return; used in NDX/SPX relative performance ratio.
97	MXWO_log_ret_21d	$\ln(\mathrm{MXWO}_t/\mathrm{MXWO}_{t-21})$	1-month MSCI World return; used in global vs. U.S. divergence signal.
98	NDX_GK_vol_21d	Garman-Klass volatility, Nasdaq, 21-day window	Nasdaq-specific volatility; elevated Nasdaq vol relative to broad market suggests concentration risk in technology.
99	MXWO_GK_vol_21d	Garman-Klass volatility, MSCI World, 21-day window	Global equity volatility; elevated alongside US stress confirms systemic rather than idiosyncratic risk.

Table 2: Bloomberg Engineered Features (continued)

#	Feature Name	Computation / Formula	Rationale
100	NDX_SPX_ratio	NDX / SPX_PX_LAST	Level of Nasdaq relative to S&P 500; rising ratio indicates market breadth is narrowing into tech concentration.
101	NDX_SPX_ratio_chg_21d	21-day pct_change(NDX/SPX)	Rising Nasdaq-to-SPX ratio change signals market concentration risk — narrow breadth historically precedes fragile market states.
102	corr_SPX_NDX_21d	Rolling 21-day Pearson correlation of SPX and NDX daily returns	Falling correlation between indices that normally co-move signals breadth fragmentation and early-stage stress.
103	corr_SPX_MXWO_21d	Rolling 21-day Pearson correlation of SPX and MXWO daily returns	Falling global-US equity correlation indicates divergence between U.S. markets and the rest of the world, a sign of idiosyncratic U.S. risk.
104	index_return_dispersion_21d	Std dev of 21-day returns across SPX, DJIA, NDX, MXWO	Rising return dispersion across major indices reflects market disagreement — associated with regime transitions and fragile market states.
105	MXWO_SPX_ret_divergence_21d	MXWO_log_ret_21d − SPX_log_ret_21d	Direct global vs. U.S. performance gap. U.S. underperformance relative to global equities signals U.S.-specific stress; global underperformance signals systemic risk.
K. Multi-Asset Composite Stress Score			
106	stress_vix_spike	1 if VIX_ret_5d > 0.20, else 0	Binary flag activated when VIX rises more than 20% in 5 days — a threshold associated with acute fear escalation.
107	stress_gold_outperform	1 if GOLD_rel_strength_21d > 0.03, else 0	Binary safe-haven flight flag; activated when gold outperforms SPX by more than 3% over a month.
108	stress_yc_steepening	1 if YC_10Y_2Y_chg_10d > 0.20, else 0	Binary de-inversion flag; rapid curve steepening of 20+ bps in 10 days signals imminent economic deterioration risk.
109	stress_oil_spike	1 if OIL_log_ret_21d > 0.15, else 0	Binary flag for supply shock; oil rising 15%+ in a month is the threshold associated with recession risk in historical episodes.
110	stress_vol_regime	1 if SPX_vol_ratio_5_63 > 1.50, else 0	Binary volatility regime flag; short-term vol exceeding long-term by 50% signals a volatility regime transition.
111	stress_composite_score	Sum of the 5 binary stress flags above (range: 0–5)	Multi-condition stress indicator. Empirically validated: drawdown rate rises from 36.2% at score 0 to 56.7% at score 4, confirming monotonic signal strength.

Table 3. Reddit Dataset from HuggingFace API

Col	Social Media Field	Meaning
A	title	The headline of the Reddit post. This is often where the specific stock ticker (e.g., $AAPL, $TSLA) is mentioned.
B	text	The main content of the post.
C	created_datetime	The exact time the post was published.
D	score	The net number of upvotes. Mathematically, it is Upvotes minus Downvotes.
E	num_comments	The total number of comments on that post. High comment counts often indicate a "hot" or controversial topic.
F	upvote_ratio	A decimal between 0 and 1 (or a percentage) representing how many people upvoted vs. downvoted. Note: A 0.95 ratio means 95% of people liked the post, indicating high community agreement.
G	flair	A "tag" assigned by the user or moderator (e.g., "DD" for Due Diligence, "News," or "Technical Analysis"). This helps you filter for high-quality research versus memes.
H	subreddit	Which community the post came from (e.g., r/stocks, r/wallstreetbets, or r/investing).
I	company	Name of the corporation associated with the stock mentioned in the post.

Table 6: Social Media Dataset (predictions.csv)

Col	Social Media Field	Meaning
A	id	The unique alphanumeric identifier for the Reddit post (useful for de-duplication).
B	title	The headline of the Reddit post. This is often where the specific stock ticker (e.g., $AAPL, $TSLA) is mentioned.
C	text	The main content of the post.
D	created_utc	The timestamp of the post in Unix format (seconds since 1970).
E	created_datetime	The exact time the post was published.
F	author	The Reddit username of the person who created the post.
G	score	The net "Upvotes" minus "Downvotes." This represents how much the community agreed with or liked the post.
H	num_comments	The total number of comments on that post. High comment counts often indicate a "hot" or controversial topic.
I	upvote_ratio	A decimal between 0 and 1 (or a percentage) representing how many people upvoted vs. downvoted. Note: A 0.95 ratio means 95% of people liked the post, indicating high community agreement.
J	flair	A "tag" assigned by the user or moderator (e.g., "DD" for Due Diligence, "News," or "Technical Analysis"). This helps you filter for high-quality research versus memes.
K	permalink	Direct links to the original post on Reddit.
L	url	Direct links to the original post on Reddit.
M	subreddit	Which community the post came from (e.g., r/stocks, r/wallstreetbets, or r/investing).
N	company	Name of the corporation associated with the stock mentioned in the post.
O	sentiment	The probability score generated by DistilBERT model. Values closer to 1.0 indicate high confidence that the post is Bullish/Positive. Values closer to 0.0 indicate high confidence that the post is Bearish/Negative.
P	date	A cleaned version of the created_datetime (likely showing just the day), used for the final aggregation into your daily trends.

Table 7: Social Media Dataset (daily.csv)

Col	Social Media Field	Meaning
A	date	The specific calendar day for which the sentiment was aggregated (formatted as DD/MM/YYYY). This serves as the Primary Key or temporal index. It allows you to align the Reddit "Social Signal" with the daily "Close" prices from the S&P 500 and the macroeconomic data from Bloomberg.
B	sentiment	The Weighted Mean Probability Score of all Reddit posts processed on that specific day. Technical Scale: This is a continuous probabilistic variable ranging from 0.0 to 1.0. • $x > 0.5$: Indicates a "Bullish" or positive daily outlook. • $x < 0.5$: Indicates a "Bearish" or negative daily outlook. • $x \approx 0.5$: Indicates a "Neutral" market sentiment.
C	post_count	The total volume (frequency) of valid, ticker-verified posts identified on that day. This measures Social Volume and serves as a proxy for "Market Attention." • High post_count + High sentiment = Strong Bullish Consensus. • High post_count + Low sentiment = Panic/Fear Regime.

Table 8: Social Media Dataset (daily_sentiment_results.csv)

Col	Social Media Field	Meaning
A	date	The primary temporal index (YYYY-MM-DD). This is used as the joining key to merge Reddit data with S&P 500 price data and Bloomberg terminal exports.
B	weighted_sentiment	The daily average sentiment score, where individual posts are weighted by their community engagement (upvote ratio). It ranges from 0 (extreme bearishness) to 1 (extreme bullishness), with 0.5 representing market neutrality.
C	total_mentions	The absolute volume of posts identified for that day. This serves as a proxy for Social Liquidity or market attention; high volume suggests a stronger consensus behind the day's sentiment score.
D	num_tickers	The count of unique S&P 500 companies discussed that day. This helps distinguish between a "Broad Market Move" (many tickers mentioned) and a "Single Stock Event" (only one or two tickers mentioned).
E	sentiment_ma7	The 7-day Simple Moving Average. This captures the weekly "buzz" or retail hype cycle, effectively filtering out daily outliers to show the short-term trend.
F	sentiment_ma30	The 30-day Simple Moving Average. This represents the long-term sentiment baseline, often reflecting more institutional or fundamental shifts in market perception rather than temporary social media spikes.
G	momentum	Calculated as the divergence between the short-term and long-term trends (MA7 − MA30). • Positive Momentum: Indicates retail enthusiasm is accelerating faster than the monthly trend (Bullish signal). • Negative Momentum: Suggests the social mood is cooling off or turning fearful (Bearish signal).
H	week	The week of the calendar month.
I	month	The month of the calendar year.

Table 9: Social Media Dataset (test_predictions.csv)

Col	Social Media Field	Meaning
A	date	The specific calendar day the post was authored. This is the primary key used to join the sentiment signal with Bloomberg macroeconomic data and S&P 500 price series.
B	pred_label	The final classification assigned by the DistilBERT model. • 1: Bullish/Positive sentiment. • 0: Bearish/Negative sentiment.
C	pred_prob	The "Softmax" probability score (0.0 to 1.0). It represents the model's mathematical confidence; values near 1.0 indicate extreme bullishness, while values near 0.5 suggest the text was ambiguous or neutral.
D	title	The headline of the Reddit post. In the preprocessing stage, this text was tripled in the input string to simulate higher cognitive importance for the Transformer's attention mechanism.
E	tickers	A list of S&P 500 constituent symbols identified within the text via Regex. This serves as the Asset Link, connecting qualitative opinions to specific companies in the main financial dataset.

7.2 Models/Algorithms

1. Linear and Statistical Models

a. Logistic Regression

Logistic Regression is one of the standard models for binary classification. It computes a weighted sum of the input features, then passes the sum through a sigmoid function to convert it into a probability between 0 and 1. The learned coefficients indicate how strongly each feature pushes the prediction toward the downturn class or the non-downturn class. The model is linear in feature space, creating a single global decision boundary. Its main strength is interpretability. Each coefficient has a clear directional meaning, and the model is relatively stable when the feature space is not too noisy. It is also computationally efficient and serves as a good first check of whether the signal in the data is approximately linear.

Why test it?

In the financial dataset, many features represent market conditions such as volatility, momentum, and trend, and it is natural to first test whether a weighted combination of those indicators is already sufficient to separate downturn and non-downturn periods. By evaluating the performance of Logistic Regression, we can determine whether it is necessary to implement complex non-linear modelling on the problem.

b. Linear Discriminant Analysis (LDA)

LDA is both a classifier and dimensionality-reduction method, it assumes that each class is generated from a multivariate Gaussian distribution and both classes share the same covariance matrix. Instead of fitting coefficients directly like LogReg, LDA estimates the class means and the common covariance matrix, then determines the projection that best separates the classes relative to their within-class spread.

Why test it?

The engineered financial indicators may form broad market regimes where downturn periods have a shifted center relative to normal periods. LDA is a useful benchmark for that type of structure because it checks whether the classes differ mainly in their average location in feature space rather than through complicated local nonlinearities.

c. Quadratic Discriminant Analysis (QDA)

QDA extends LDA by allowing each class to have its own covariance matrix, implying the model no longer assumes the downturn and normal periods have the same geometric spread in feature space, therefore the decision boundary becomes quadratic/curved. It is more flexible than LDA, able to adapt to dispersed classes by fitting a separate covariance matrix.

Why test it?

Financial stress periods often look more volatile and dispersed than normal periods. QDA is relevant because downturn days may not only shift in average feature values, but also differ in variance structure.

d. Gaussian Naive Bayes (GNB)

While GNB is extremely fast and simple, it also maintains a good performance when the independent marginal signals are informative. It treats each feature as a separate source of evidence and combines them probabilistically.

Why test it?

Although the independence assumption is unrealistic for financial indicators, GNB is useful as a baseline to test whether downturn prediction is already visible in the marginal behavior of the features.

2. Kernel and Distance-Based Models

a. Support Vector Machine (SVM)

SVM looks for the decision boundary that maximizes the margin between two classes. In the non-linear case, a kernel function maps the data into higher-dimensional space where a linear separation becomes possible. With a nonlinear kernel, SVM produced highly flexible boundaries while maintaining a strong generalization.

Why test it?

In the financial dataset, the downturn risk emerges from combinations such as volatility together with weakening momentum and broader stress signals. SVM can represent nonlinear boundaries without requiring manual interaction engineering.

b. K-Nearest Neighbors (KNN)

To classify a sample, KNN finds the nearest training points (Euclidean distance) in feature space and assigns the class based on the majority label among those neighbors. KNN can adapt to irregular local structures without imposing a global functional form.

Why test it?

KNN may not become the strongest model, but it is informative to test the hypothesis that the data contain local neighborhood patterns rather than only smooth global relationships.

3. Tree and Ensemble Models

a. Random Forest

Random forest builds a large number of decision trees. Each tree is trained on a bootstrap sample of the training data, and at each split only a random subset of features is considered. The final prediction is made by averaging probabilities or taking a majority vote across all trees. RF is strong on tabular data with nonlinear effects and interactions. It can capture threshold behaviour automatically, such as a feature becoming informative only above a certain level.

Why test it?

The financial dataset contains engineered indicators whose relationship to future downturns is non-linear. RF can model interaction effects such as volatility, momentum and stress variables working together.

b. Extra Trees (Extremely Randomized Trees)

Extra Trees introduced more randomness. Instead of searching carefully for the best split threshold at each node, it selects split thresholds at random and chooses among them. The extra randomness can reduce variance and sometimes improve generalization, especially when the training data are noisy.

Why test it?

The financial classification problems often contain noise and unstable short-term behavior, Extra Trees check whether the dataset benefits from broader ensemble diversity rather than more carefully optimized tree splits.

c. AdaBoost

AdaBoost builds learners sequentially by starting with a simple weak learner, and increases the weight of observations that were previously misclassified after each round. Later learners therefore focus more on the difficult examples. Rather than averaging many independent trees, AdaBoost repeatedly concentrates on the observations the earlier model got wrong.

Why test it?

Pre-downturn periods may be difficult to classify because some of them initially resemble ordinary market weakness rather than true stress buildup. AdaBoost can give progressively more attention to such borderline cases.

d. Histogram Gradient Boosting (HGB)

HGB is a boosted-tree method that first bins continuous variables into histograms, then grows trees to reduce the residual errors of the current ensemble. The histogram approximation makes training faster and more scalable. HGB is strong at fitting nonlinear interactions and subtle decision boundaries.

Why test it?

The financial features are all tabular and largely numerical, HGB is appropriate for learning higher-order interactions among variables while remaining computationally manageable.

e. LightGBM

LGBM is a gradient boosting framework specialized for efficient tree boosting. It grows tree leaf-wise rather than level-wise by expanding the leaf that most improves the loss at each step. It is designed for speed, memory efficiency and strong performance on structured tabular data. It often handles feature interactions effectively and is widely used in competitive structured-data tasks.

Why test it?

LGBM may perform well on the pure financial dataset due to the structured numeric predictors, possible nonlinear thresholds, and complex cross-feature interactions.

f. XGBoost

XGB is a gradient-boosted tree framework. It builds trees sequentially, each one aimed at correcting the residual error of the ensemble, while adding regularization terms to control overfitting. It also incorporates shrinkage, subsampling and optimized training procedures. XGB often performs well when the signal depends on multiple interacting features and when careful control of model complexity matters.

Why test it?

XGB tests whether a highly optimized boosting algorithm can separate downturn and non-downturn periods more effectively given a structured and nonlinear financial feature space.

g. CatBoost

CatBoost is a gradient boosting method which includes training mechanisms designed to reduce target leakage and prediction shift. It is highly competitive on all-numeric tabular datasets. It constructs boosted decision trees while using ordered boosting ideas for better generalization.

Why test it?

The financial dataset is tabular and interaction-heavy, by implementing CatBoost helps ensure the screening stage is not biased toward only one boosting implementation, given different boosting frameworks applied.

4. Deep Learning Models

a. MLP Neural Network

A Multilayer Perceptron consists of several fully connected layers, where each layer applies a weighted transformation followed by a nonlinear activation function. As information moves through the network, the model learns increasingly abstract transformations of the original features. The final layer outputs a probability for the downturn class.

Why test it?

The financial indicators may interact in ways that are difficult to hand-specify, and an MLP can learn those combinations directly. It is therefore a reasonable candidate for testing whether a general nonlinear function approximator can extract more signal from the tabular features.

b. TabNet

TabNet is a deep neural architecture designed for tabular data. Instead of processing all features in the same way at every step, it uses sequential attention masks to decide which features to focus on at each decision stage. In effect, the model performs a series of learned feature-selection and transformation steps before producing the final output. TabNet combines neural-network flexibility with a structure tailored to tabular inputs. Its attentive mechanism gives it a kind of dynamic feature selection, allowing different observations to emphasize different subsets of features.

Why test it?

TabNet is one of the deep architectures specifically designed for classical tabular financial datasets. In principle, it is worth testing to learn richer tabular interactions while also adapting which features matter most across different market states.

5. Stacked Model — Soft Voting Ensemble

A soft voting ensemble combines multiple base models by averaging their predicted probabilities for the positive class, then using the aggregated probability for the final prediction. Different models often make different errors because they capture different aspects of the data. Soft voting ensemble reduces reliance on any single model and can smooth out unstable predictions.

Why test it?

The financial dataset may contain a mixture of linear, threshold-based, and interaction-driven signals. A voting ensemble can pool these different modelling biases into one probability score.

8. References

Dichtl, H., Drobetz, W., Liu, B., Reibnegger, K., & Wendt, S. (2023). Forecasting stock market crashes via machine learning. *Finance Research Letters*, 53, 103678.

Sheng, Y., Zhang, Y., Li, P., & Wang, J. (2024). Stock price crash prediction based on multimodal data machine learning framework. *Finance Research Letters*.

Ting, K. M., & Witten, I. H. (1997). Stacked generalization: When does it work? *Proceedings of the Fifteenth International Joint Conference on Artificial Intelligence*, 866–871.

Wolpert, D. H. (1992). Stacked generalization. *Neural Networks*, 5(2), 241–259.

9. Sources

DataHub. (n.d.). S&P 500 companies dataset (constituents.csv). Retrieved April 17, 2026, from https://datahub.io/core/s-and-p-500-companies/r/constituents.csv

Datasets. (n.d.). S&P 500 companies: Constituents dataset. GitHub. Retrieved April 17, 2026, from https://raw.githubusercontent.com/datasets/s-and-p-500-companies/main/data/constituents.csv

Emilpartow. (n.d.). Reddit finance posts S&P 500 dataset. Hugging Face. Retrieved April 17, 2026, from https://huggingface.co/datasets/emilpartow/reddit_finance_posts_sp500